

DICOT ROOT VS. MONOCOT ROOT

Feature	Dicot Root	Monocot Root
Epidermis	Single-layered, thin-walled cells; root hairs abundant for absorption	Same as dicot; root hairs present but often shorter-lived
Cuticle	Absent	Absent
Hypodermis	Usually absent	Usually absent
Cortex	Many layers of parenchyma, loosely packed; intercellular spaces allow gas diffusion	Many layers of parenchyma; often larger and with more prominent air spaces
Endodermis	Single layer; well-developed Casparian strips	Single layer; Casparian strips present
Pericycle	Single - multilayered; origin of lateral roots and vascular cambium	Single-layered; lateral roots arise; secondary growth absent

Vascular Cambium	Present → allows secondary thickening	Absent → no secondary growth
Pith	Inconspicuous or small	Large and prominent, well-developed
Xylem Arrangement	Radial; diarch, triarch, tetrach, pentarch, hexarch) common	Radial; polyarch, typically >6 arms
Phloem Arrangement	Alternates with xylem; equal number of groups	Alternates with xylem; usually more numerous
Conjunctive Tissue	Present, Parenchymatous	Present, Parenchymatous
Secondary Growth	Present due to vascular cambium → formation of secondary xylem & phloem	Absent
Stele Type	Protostele or actinostele	Polyarch protostele
Root Hairs	Numerous; increase absorption	Numerous, but often shorter lived

Diagram

Dicot root

(Cross-section)

Monocot root

(Cross-section)

